Abstract Linear Algebra

Prof. Lanier
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Pick up a handout. Try those warm-up problems.

Next up:

- This week: Chapter 2.4-8 of *LADW*
- Quiz 3 on Friday.
- First requizzing session today (4-5pm Eck 207A). Also OH.
- Homework 2 is due at the end of the week.
- Problem session Friday 3-5pm in Ryerson 177. (Possibly snacks.)

Matrix Multiplication

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 5 \\ 6 & 7 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 15 & 20 \\ 7 & 8 \\ 23 & 32 \\ 9 & 13 \end{bmatrix}$$

Reset

Algebra Geometry Pre-Calculus Calculus Linear Algebra Discrete Math Probability/Statistics Linear Programming

Home / Calculators / Calculators: Linear Algebra / Linear Algebra Calculator

Reduced Row Echelon Form (RREF) Calculator

Find reduced row echelon form step by step

The calculator will find the row echelon form (simple or reduced – RREF) of the given (augmented if needed) matrix, with steps shown.

Related calculators: Gauss-Jordan Elimination Calculator, Matrix Inverse Calculator

Size of the matrix: 2 × 3

Matrix:

1	5	1
2	11	5

WolframAlpha computational intelligence.

Enter what you want to calculate or know about

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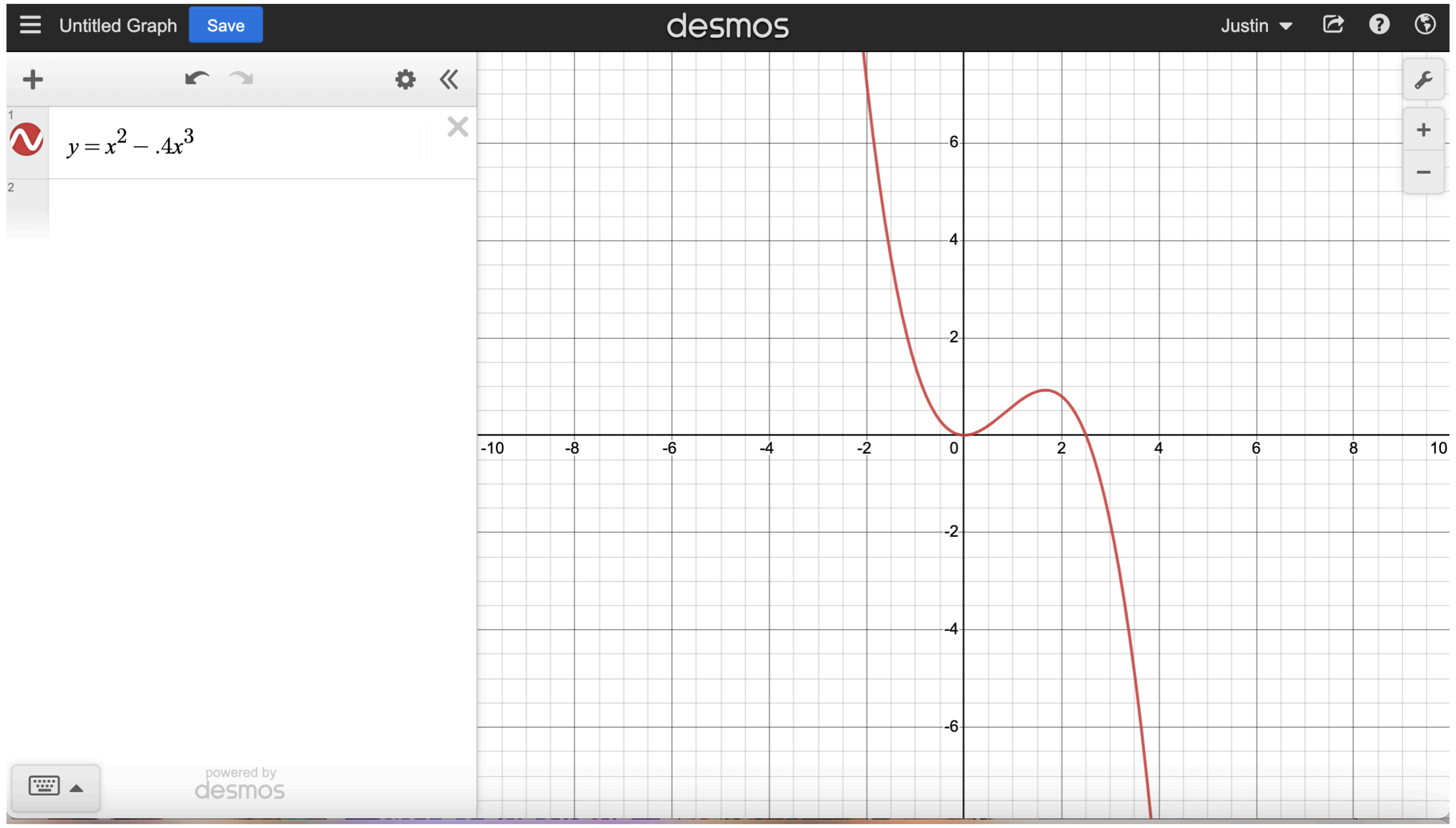

The course is structured for the exams to be tough; they are the only component of the course where it is not straightforward to get a near-perfect score.

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they are the only component of the course where it is
not straightforward to get a near-perfect score.

The exams are tough but fair.
I will give you a lot of helpful guidance about how to prepare.

Midterm exam (2021)

class average	76

Midterm exam (2021)

class average	76
average for students who attended OH at least 3 times	

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class average	76
average for students who attended OH at least 3 times	79
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Midterm exam (2021)

class average	76
average for students who attended OH at least 3 times	79
average for students who filled out a study group form at least twice	79
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average for students who laugh at my jokes in class	

Midterm exam (2021)

class average	76
average for students who attended OH at least 3 times	79
average for students who filled out a study group form at least twice	79
average for students who filled out a textbook reading form at least twice	87
average for students who laugh at my jokes in class	98

Midterm exam (2021)

class average	76
average for students who attended OH at least 3 times	79
average for students who filled out a study group form at least twice	79
average for students who filled out a textbook reading form at least twice	87
average for students who laugh at my jokes in class	98

Flex helps your grade *twice*: it gets you flex points and it helps you learn the content better.

Compute the inverse matrix for $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Three methods for computing matrix inverses (when they exist):

- use geometric intuition
- use row reduction
- use a formula

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} =$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} =$$

$$\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} =$$

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$$\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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$$\left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right)$$

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$$\left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right) \xrightarrow{R1 \leftrightarrow R2} \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array}\right)$$

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$$\left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right) \xleftrightarrow{R1 \leftrightarrow R2} \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array}\right) \xleftrightarrow{R2 - 2R1} \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array}\right)$$

$$\xleftrightarrow{R1 + R2} \left(\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & -1 & 1 & -2 \end{array}\right) \xleftrightarrow{-1R2} \left(\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array}\right)$$

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$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right]$$

{1}

A formula for the inverse of an invertible 2x2 matrix

$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right]$$

$$\boxed{\text{If the matrix is invertible, at least one of } a, c \text{ is non-zero; WLOG, assume } a \text{ is non-zero.}}$$

A formula for the inverse of an invertible 2x2 matrix

$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right] \xLeftrightarrow{R2 - \frac{c}{a}R1} \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & \frac{ad-bc}{a} & -\frac{c}{a} & 1 \end{array}\right]$$

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[1]

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ad - *bc* is non-zero;
otherwise, matrix is not invertible.

In particular,
at least one of *a*, *c* is non-zero;
WLOG, assume *a* is non-zero.

[1]

A formula for the inverse of an invertible 2x2 matrix

$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right] \overset{R2 - \frac{c}{a}R1}{\Longleftrightarrow} \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & \frac{ad-bc}{a} & -\frac{c}{a} & 1 \end{array}\right] \overset{\frac{a}{ad-bc}R2}{\Longleftrightarrow} \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & 1 & -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{array}\right]$$

$ad - bc$ is non-zero; otherwise, matrix is not invertible.

This is the determinant!

In particular, at least one of a, c is non-zero; WLOG, assume a is non-zero.

[1]

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$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right] \xLeftrightarrow{R2 - \frac{c}{a}R1} \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & \frac{ad-bc}{a} & -\frac{c}{a} & 1 \end{array}\right] \xLeftrightarrow{\frac{a}{ad-bc}R2} \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & 1 & -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{array}\right]$$

$$\Leftrightarrow \left[\begin{array}{cc|cc} a & 0 & \frac{ad}{ad-bc} & \frac{-ba}{ad-bc} \\ 0 & 1 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array}\right] \Leftrightarrow \left[\begin{array}{cc|cc} 1 & 0 & \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ 0 & 1 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array}\right]$$

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$ad - bc$ is non-zero; **This is the determinant!**
otherwise, matrix is not invertible.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

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$$\Leftrightarrow \begin{bmatrix} a & 0 & | & \frac{ad}{ad-bc} & \frac{-ba}{ad-bc} \\ 0 & 1 & | & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix} \Leftrightarrow \begin{bmatrix} 1 & 0 & | & \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ 0 & 1 & | & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$$

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$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

[1]

A formula for the inverse of an invertible 2x2 matrix

Warm-up 1

Give three different examples of matrices in $M^{\mathbb{R}}_{2\times 2}$ that are in reduced row echelon form. In each one, circle each of the pivots.

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$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & * \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Warm-up 1

Give three different examples of matrices in $M_{2\times 2}^{\mathbb{R}}$ that are in reduced row echelon form. In each one, circle each of the pivots.

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & * \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

For $M_{2\times 2}^{\mathbb{Z}/5\mathbb{Z}}$, there are exactly 8 matrices that are in RREF.

Row equivalence: matrices "connected" by row operations

$$\begin{pmatrix} 3 & 1 \\ 3 & 6 \end{pmatrix} \leftrightarrow \begin{pmatrix} 3 & 1 \\ 0 & 5 \end{pmatrix}$$

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in general,
equivalence relations are:

- reflexive
- symmetric, and
- transitive

Row equivalence: matrices "connected" by row operations

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Theorem: For every $A \in M_{m\times n}^{\mathbb{F}}$ there exists a unique $B \in M_{m\times n}^{\mathbb{F}}$ in RREF such that A is row equivalent B.

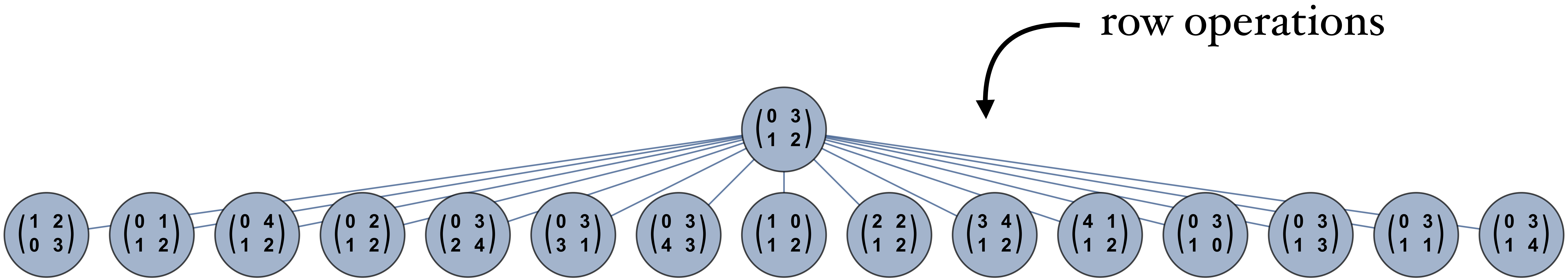
in general,
equivalence relations are:

- reflexive
- symmetric, and
- transitive

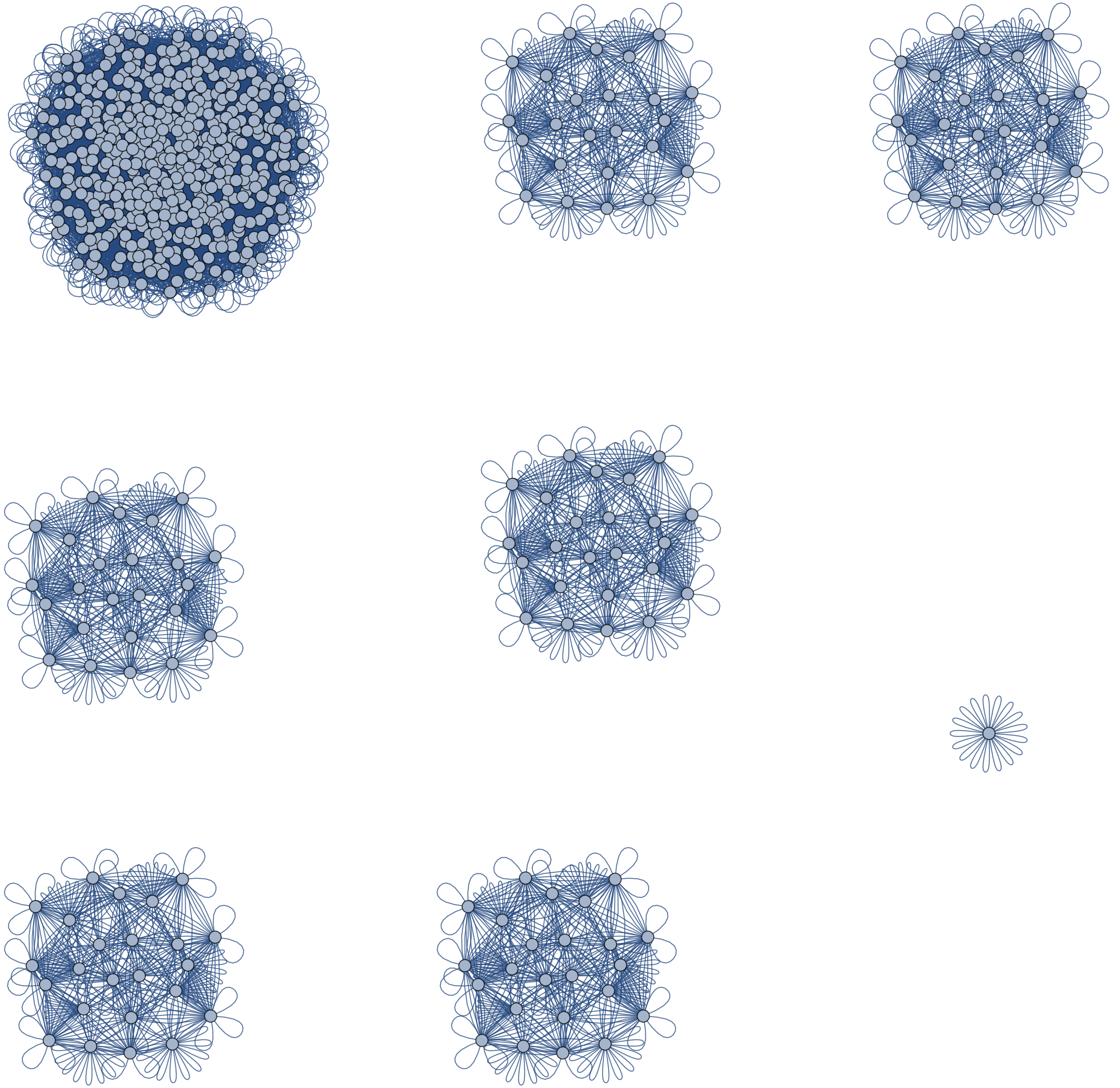
existence and uniqueness

Row reduction:
process vs. algorithm

In $M_{2\times 2}^{\mathbb{Z}/5\mathbb{Z}}$:

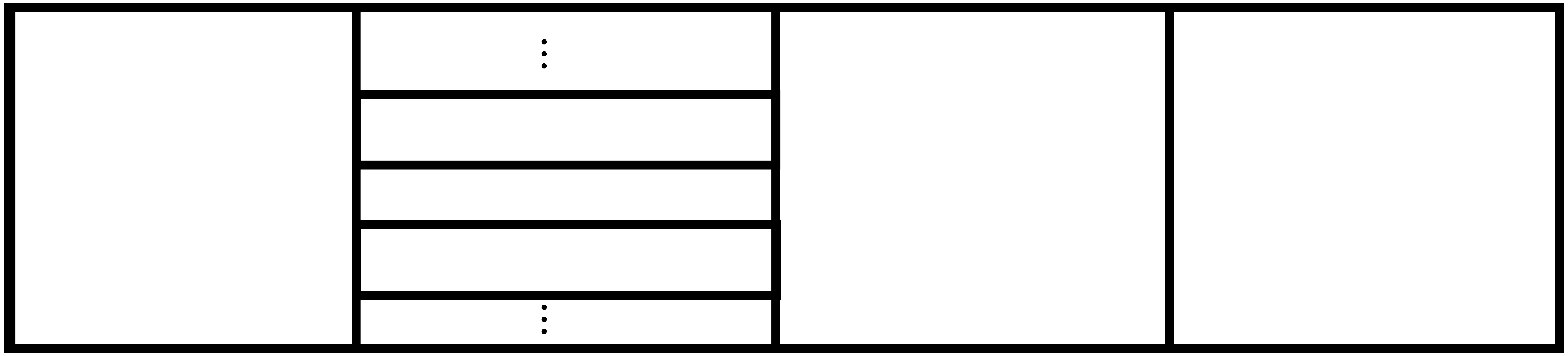


In $M_{2\times 2}^{\mathbb{Z}/5\mathbb{Z}}$,
these are the
row equivalence
classes



In $M^{\mathbb{R}}_{2\times 2}$,
these are the row equivalence classes

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & * \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$



uncountably many,
one for each real number

They exactly correspond to solution sets
of systems of two equations in two variables .

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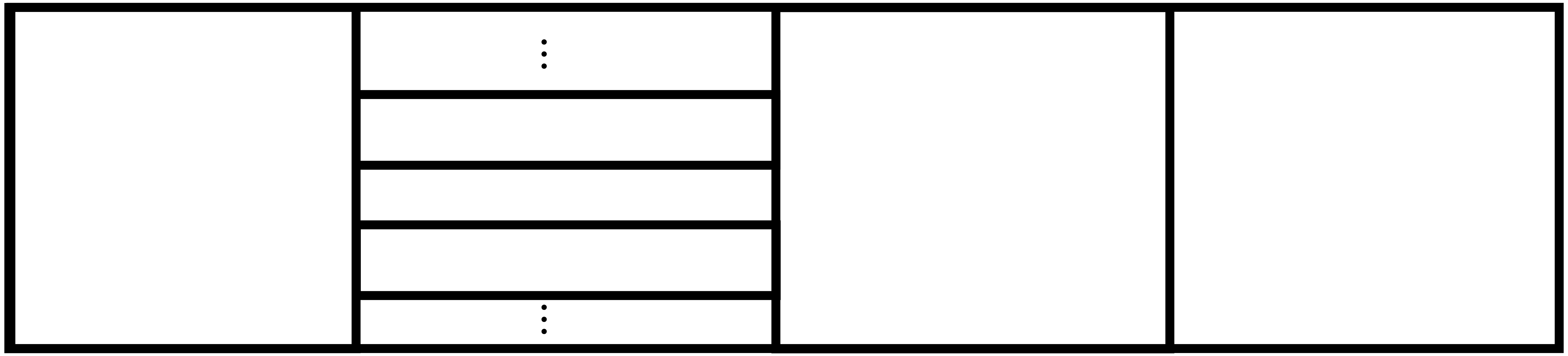

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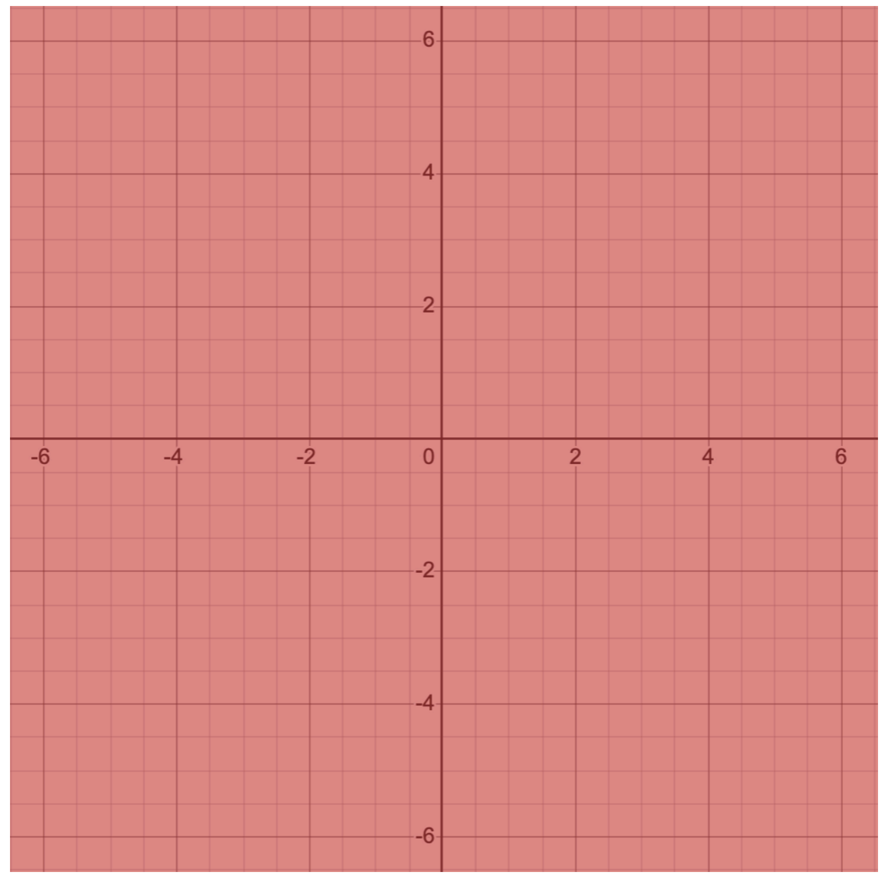
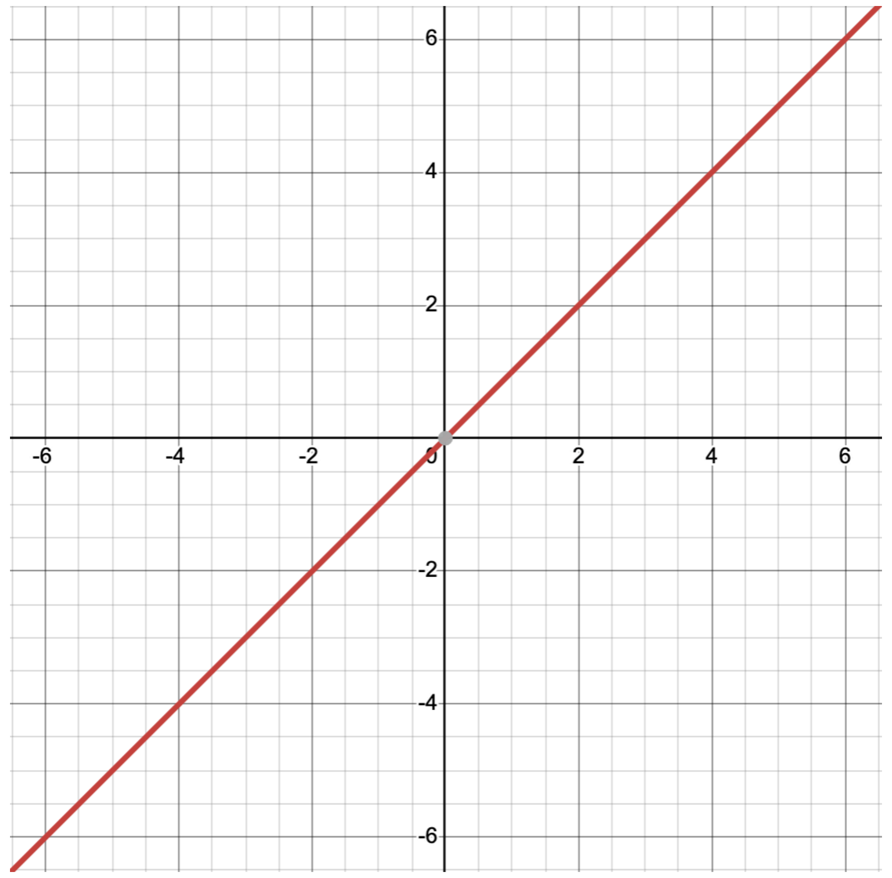
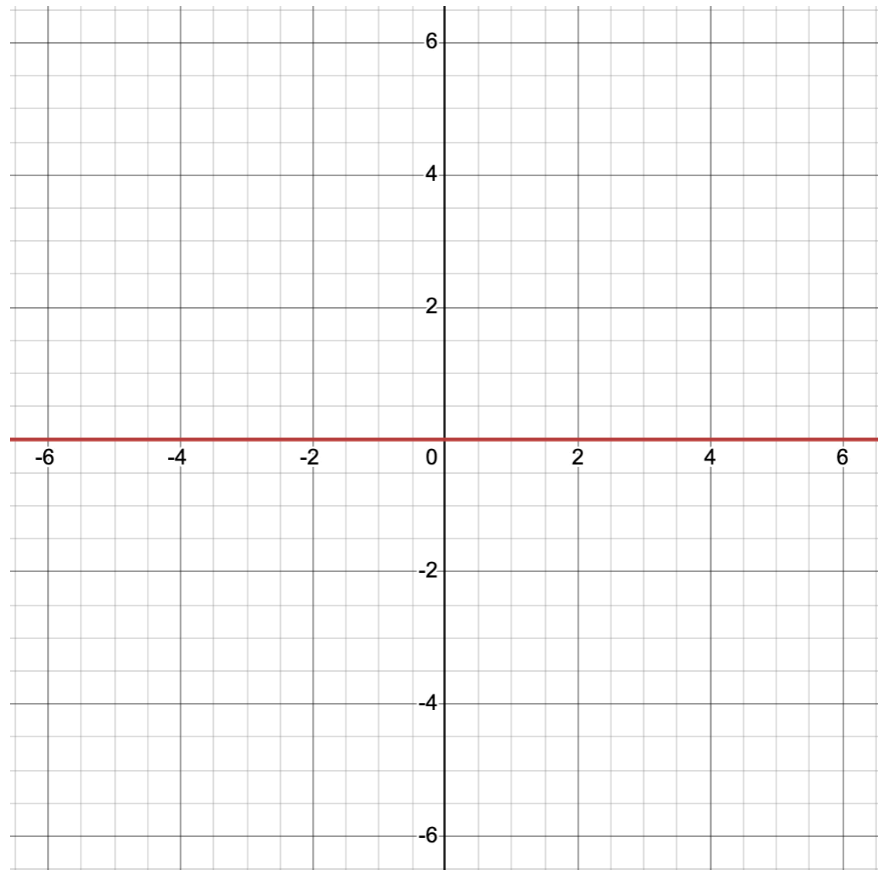
$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & * \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

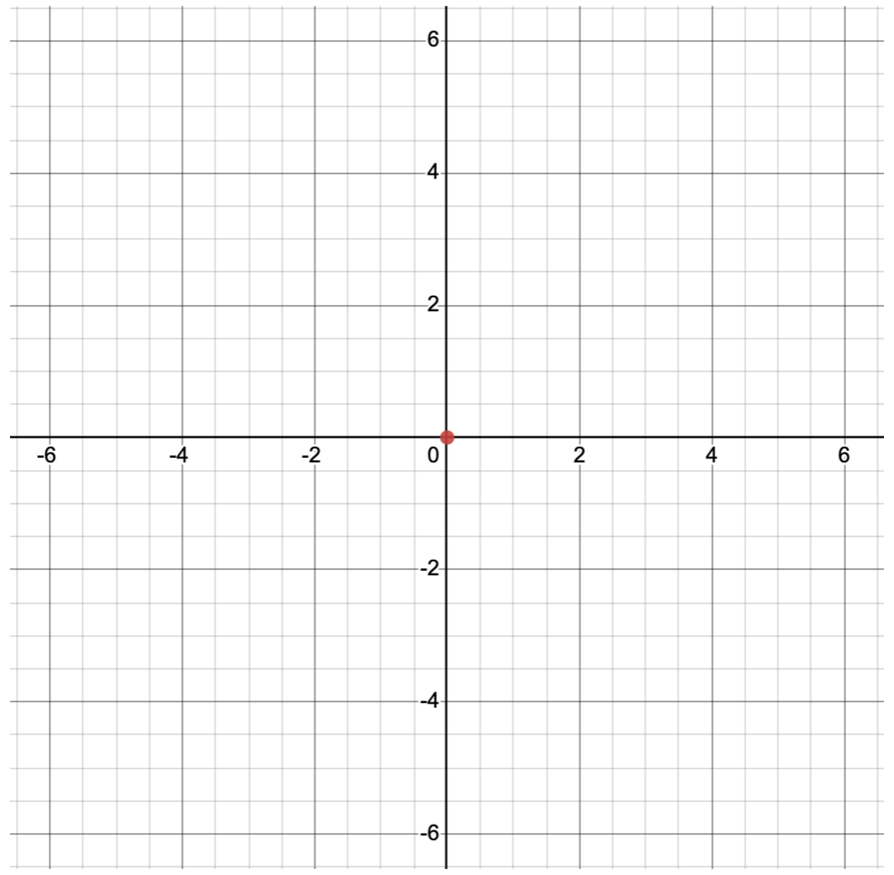
$$\begin{aligned} 0 &= 0 \\ 0 &= 0 \end{aligned} \qquad \begin{aligned} x + * \, y &= 0 \\ 0 &= 0 \end{aligned} \qquad \begin{aligned} y &= 0 \\ 0 &= 0 \end{aligned} \qquad \begin{aligned} x &= 0 \\ y &= 0 \end{aligned}$$



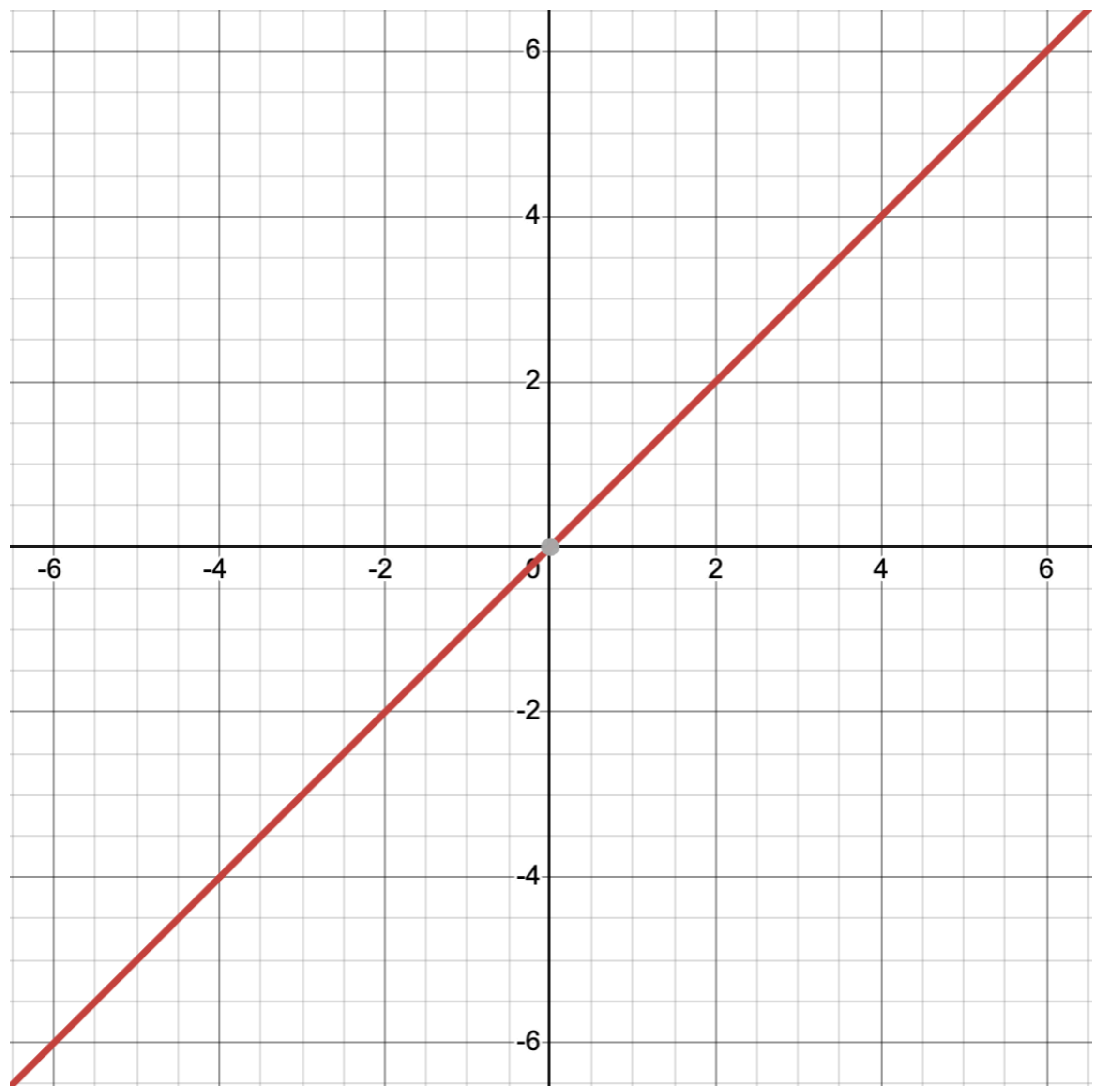


e.g.





$$x +^* y = 0$$
$$0 = 0$$



6
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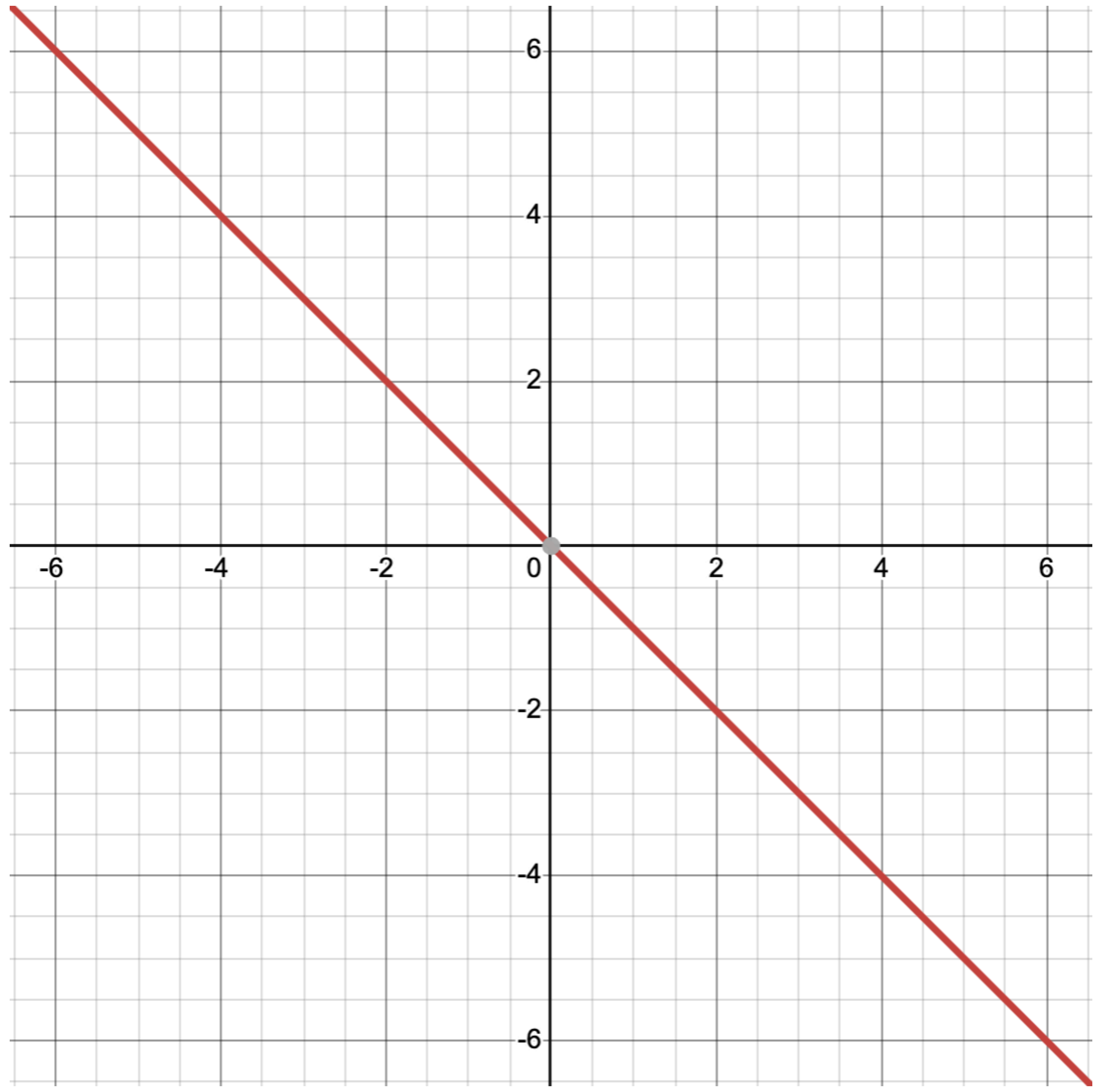

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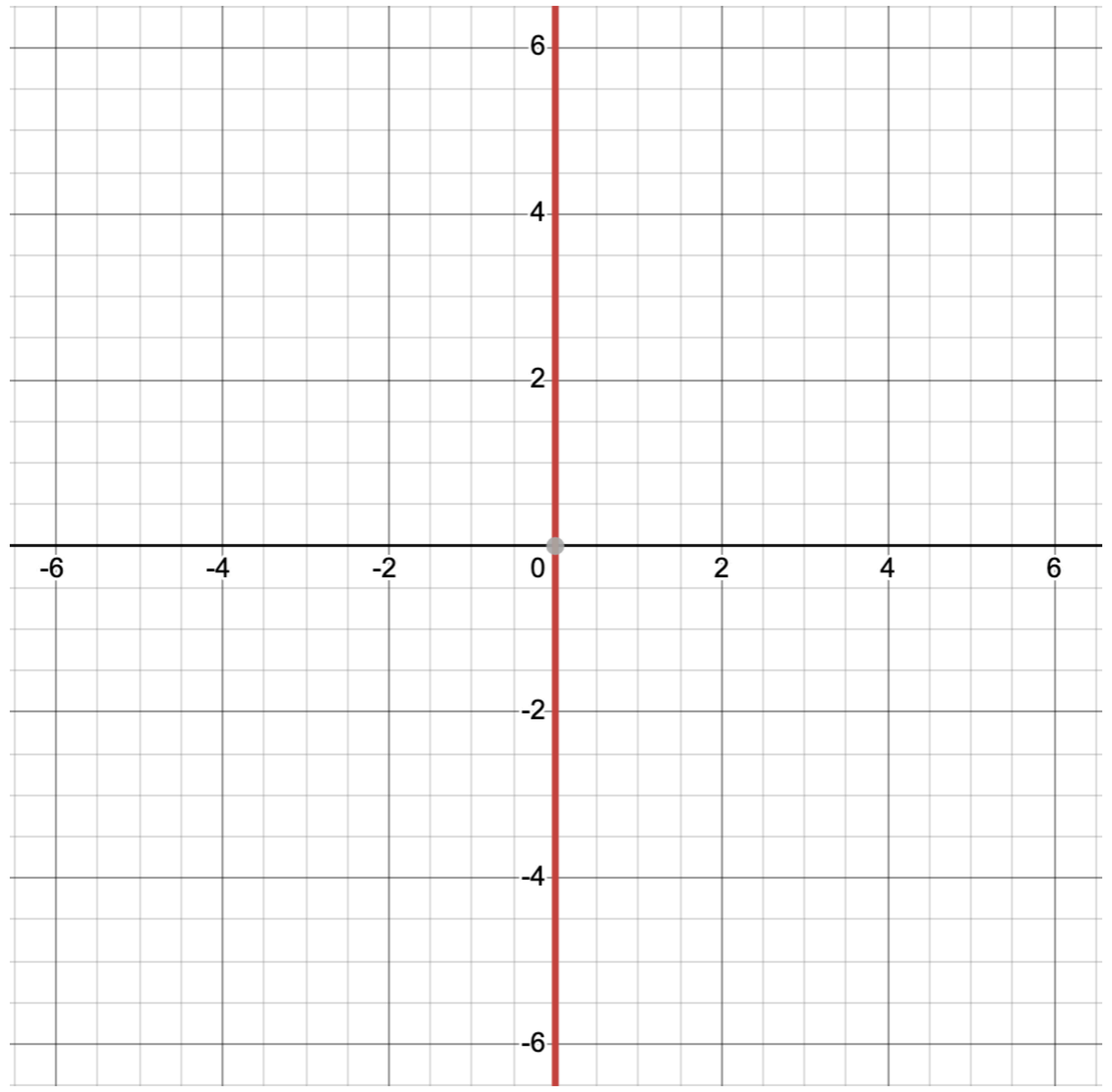

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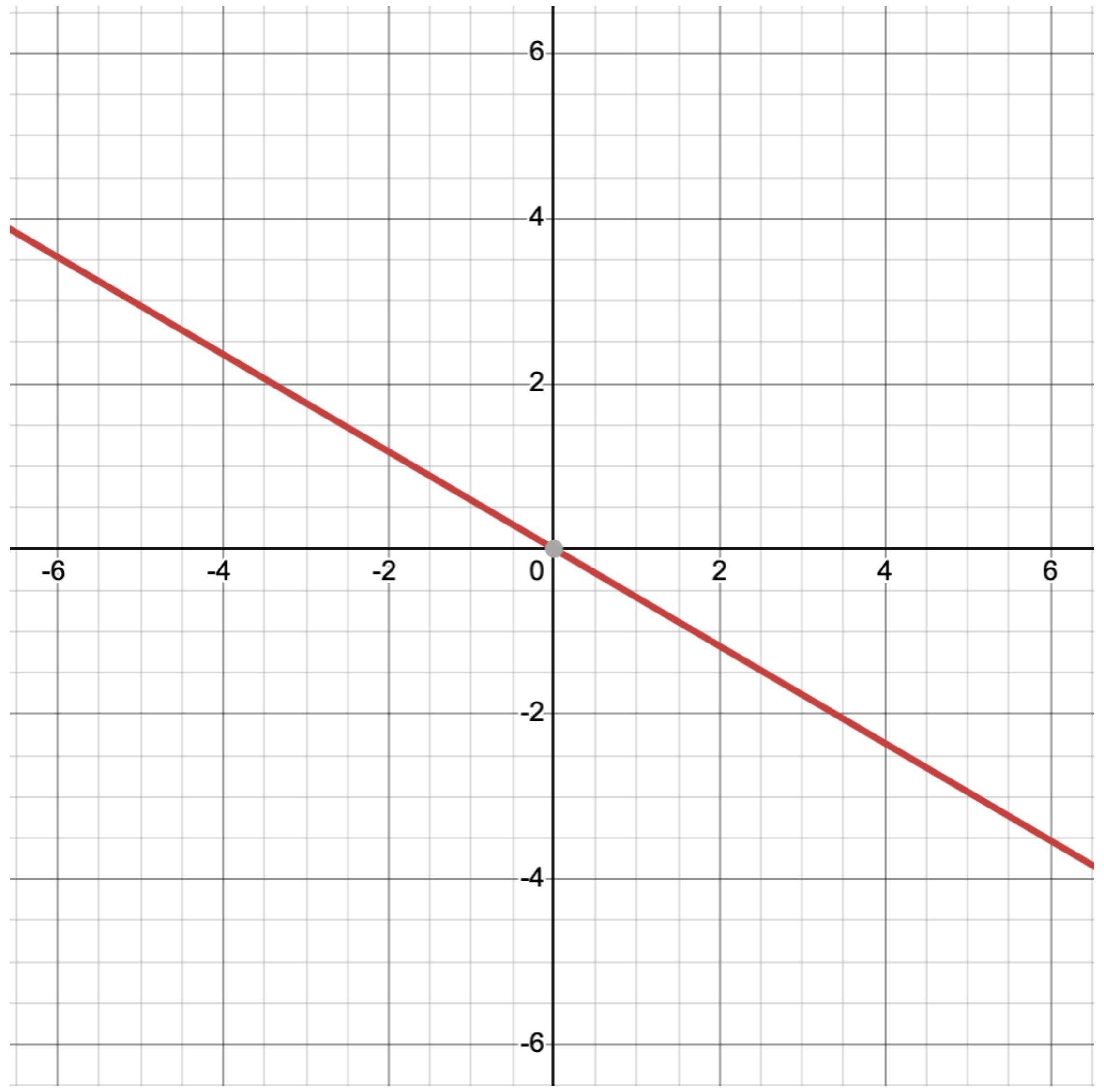

6
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In $M^{\mathbb{R}}_{2\times 2}$, these are the row equivalence classes

They exactly correspond to solution sets of systems of two equations in two variables .

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & * \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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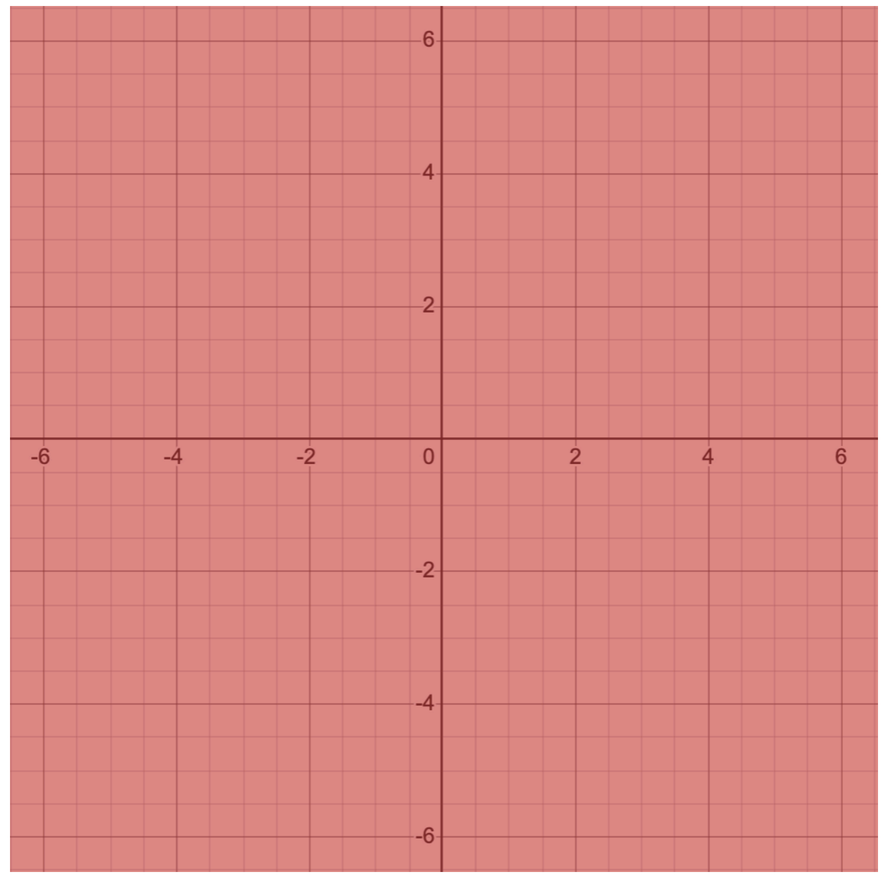

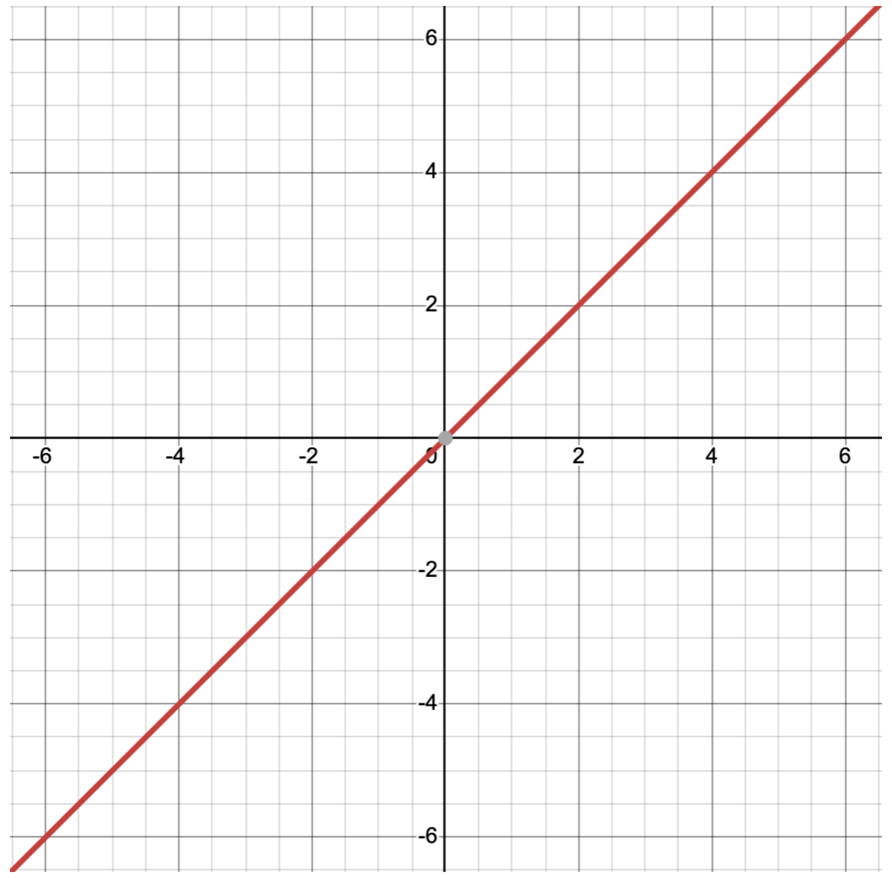

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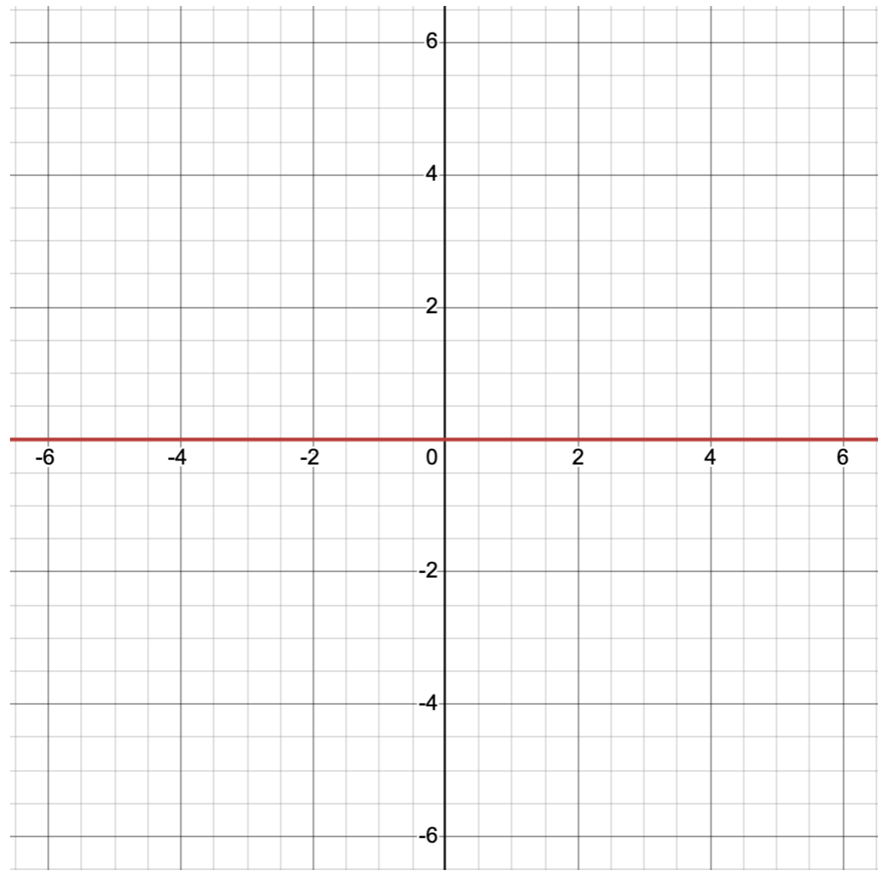

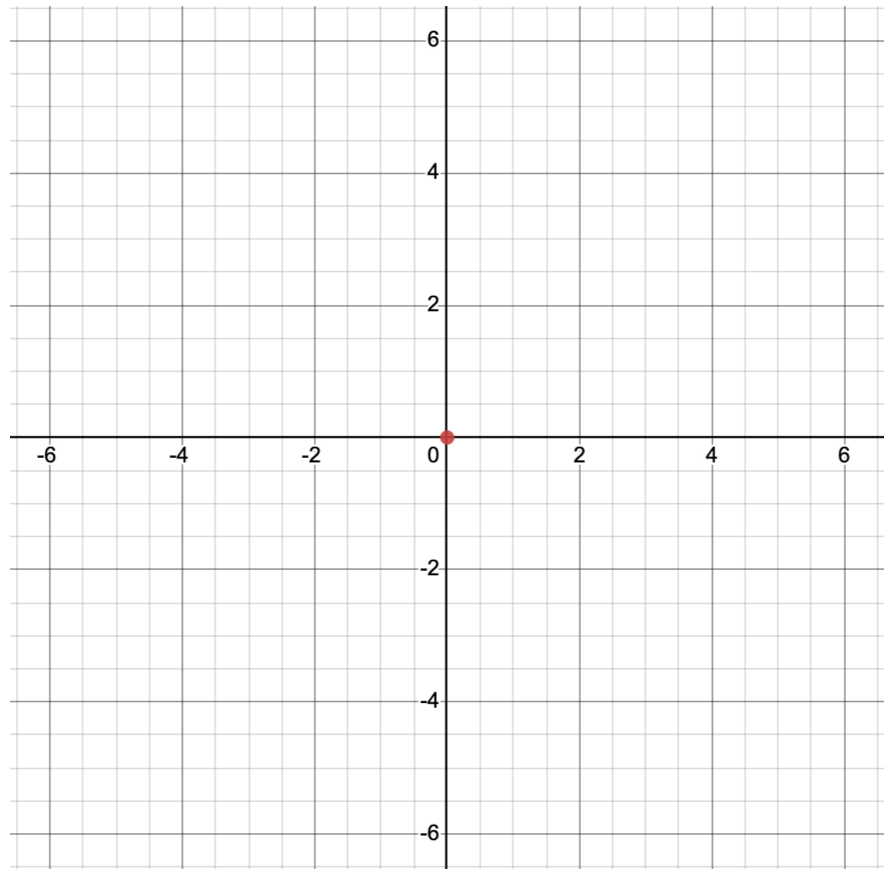

Matrix in RREF	Corresponding homogeneous system	Expression for each variable	Solution set in vector form
$\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 9 \end{pmatrix}$			

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$\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 9 \end{pmatrix}$	$\begin{aligned} w + 2x \qquad - z &= 0 \\ y + 9z &= 0 \end{aligned}$		

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$\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 9 \end{pmatrix}$	$\begin{aligned} w + 2x \quad - z &= 0 \\ y + 9z &= 0 \end{aligned}$	$w = -2x + z$ $y = -9z$	

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x and z are called free variables since they can be anything.

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$\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 9 \end{pmatrix}$	$\begin{aligned} w + 2x \quad\quad - z &= 0 \\ y + 9z &= 0 \end{aligned}$	$\begin{aligned} w &= -2x + z \\ x &= x \\ y &= -9z \\ z &= z \end{aligned}$	$\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = x \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} 1 \\ 0 \\ -9 \\ 1 \end{pmatrix}$

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$$\begin{pmatrix} 1 & 0 & 8 \\ 0 & 1 & -3 \end{pmatrix}$$

Matrix in RREF	Corresponding homogeneous system	Expression for each variable	Solution set in vector form
$\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 9 \end{pmatrix}$	$\begin{aligned} w + 2x \quad - z &= 0 \\ y + 9z &= 0 \end{aligned}$	$w = -2x + z$ $x = x$ $y = -9z$ $z = z$	$\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = x \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} 1 \\ 0 \\ -9 \\ 1 \end{pmatrix}$
$\begin{pmatrix} 1 & 0 & 8 \\ 0 & 1 & -3 \end{pmatrix}$			$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = z \begin{pmatrix} -8 \\ 3 \\ 1 \end{pmatrix}$

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Next up:

- This week: Chapter 2.4-8 of *LADW*
- Quiz 3 on Friday.
- First requizzing session today (4-5pm Eck 207A). Also OH.
- Homework 2 is due at the end of the week.
- Problem session Friday 3-5pm in Ryerson 177. (Possibly snacks.)